

Statistical Analysis: Pairwise Universe Comparisons  
Qwen3-4B (20,000 documents)

| Epoch   | Comparison       | Medians      | Difference | Cliff's $\delta$ | Effect Size | p-value† | Sig. |
|---------|------------------|--------------|------------|------------------|-------------|----------|------|
| Epoch 1 | False vs True    | 2.00 vs 2.00 | +0.00      | +0.06            | Negligible  | 0.6247   | ns   |
| Epoch 1 | False vs Neutral | 2.00 vs 2.00 | +0.00      | +0.08            | Negligible  | 0.2753   | ns   |
| Epoch 1 | True vs Neutral  | 2.00 vs 2.00 | +0.00      | +0.02            | Negligible  | 1.0000   | ns   |
| Epoch 2 | False vs True    | 3.00 vs 3.00 | +0.00      | -0.01            | Negligible  | 1.0000   | ns   |
| Epoch 2 | False vs Neutral | 3.00 vs 3.00 | +0.00      | +0.05            | Negligible  | 0.9495   | ns   |
| Epoch 2 | True vs Neutral  | 3.00 vs 3.00 | +0.00      | +0.06            | Negligible  | 0.5514   | ns   |
| Epoch 4 | False vs True    | 3.00 vs 3.00 | +0.00      | +0.12            | Negligible  | 0.0324   | *    |
| Epoch 4 | False vs Neutral | 3.00 vs 3.00 | +0.00      | +0.19            | Small       | <0.0001  | ***  |
| Epoch 4 | True vs Neutral  | 3.00 vs 3.00 | +0.00      | +0.09            | Negligible  | 0.1790   | ns   |

† p-values are Bonferroni-corrected ( $\alpha = 0.05/3 = 0.017$ )  
\*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$ , ns = not significant  
Mann-Whitney U test for significance, Cliff's  $\delta$  for non-parametric effect size  
Cliff's  $\delta$ :  $|\delta| < 0.147$  = negligible,  $< 0.33$  = small,  $< 0.474$  = medium,  $\geq 0.474$  = large